

Lecture 4

Account and security Administration

CoSc2043 – Network and System Administration

Section Overview

- ◆ Users and groups
- ◆ System Accounts
- ◆ Account Management
- ◆ System Administration Access

Purposes of accounts



Audit

Access Control

Authentication

Identity and Authentication

◆ Why usernames?

- Grant access to system
- Control access to resources
- Accountability

◆ Passwords

- Prove you are who you say you are
- Often weakest link in system security

User Accounts

UserID
User's Full Name
Password
Home Directory
Groups
System Interface?

Username Selection

- ◆ Must be unique
- ◆ Max of 8 characters
- ◆ Should be in all lower-case
- ◆ Easy to remember
- ◆ Set format
 - Combination of first & last name
 - No nicknames

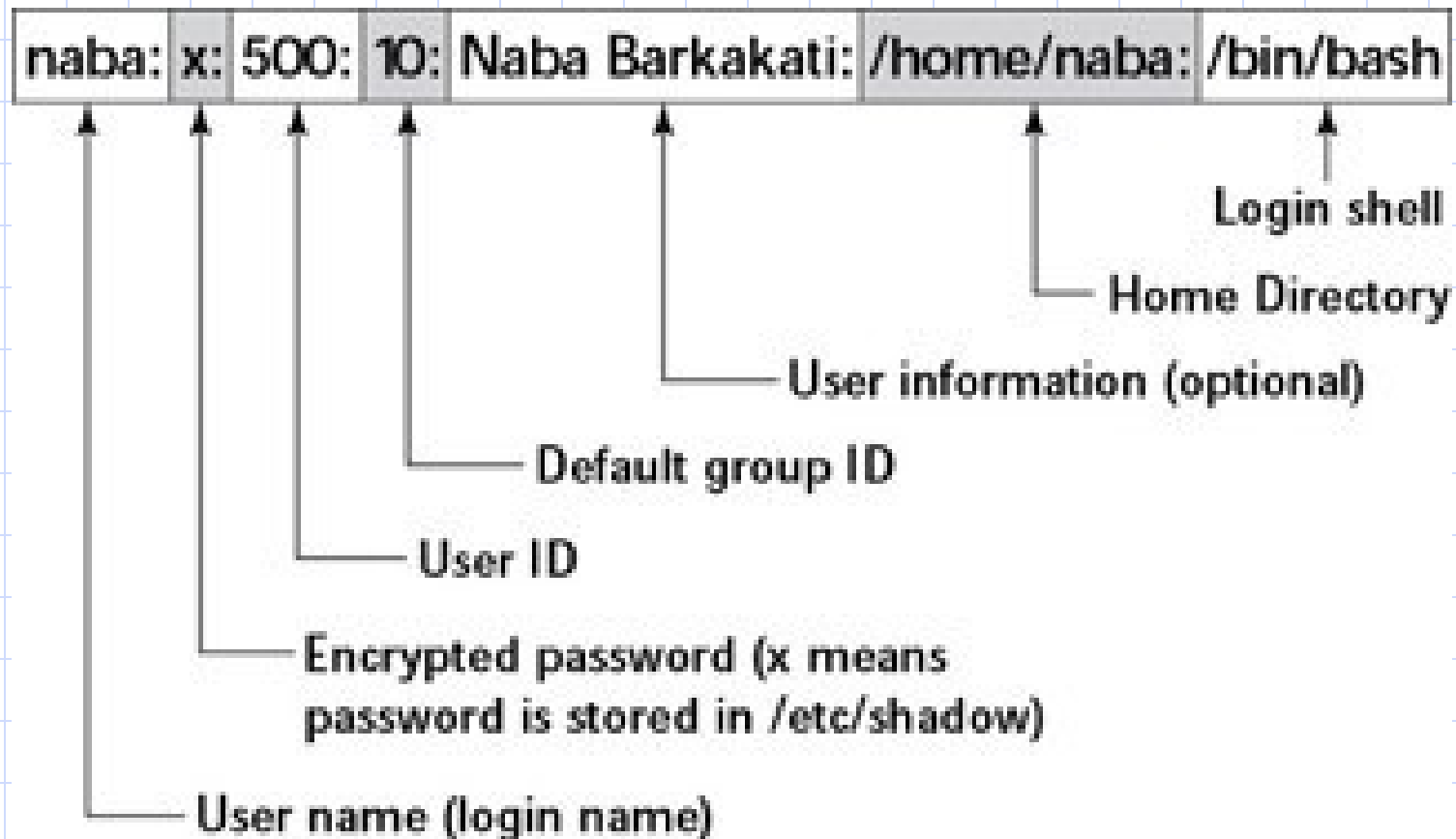
Components of an Account

- ◆ Username
- ◆ Password
- ◆ UID
- ◆ GID
- ◆ GECOS
- ◆ Home Directory
- ◆ Default Shell

◆ Stored in `/etc/passwd`

```
sorr:lYi8.KpsFAb9M:126:10:Scott Orr:/home/sorr:/bin/csh
```

Components of an Acc. . .



Microsoft Security Identifiers

- ◆ Created for every user, group, and machine
- ◆ Never reused
- ◆ S-1-5-21-D1-D2-D3-RID
 - S-1-5-21: Standard prefix for NT
 - D1-D2-D3: Local or domain identifier
 - RID (Relative ID): Unique part of SID

Weak Passwords

- ◆ No passwords used
- ◆ *Smoking Joes*
- ◆ Information about user
- ◆ Dictionary Attacks
 - Modification of user ID or name
 - Modification of dictionary(s) word(s)
 - Keyboard patterns
 - Any systematic, algorithmic generator

Selecting Strong Passwords

- ◆ At least 14 characters in length
- ◆ Mix of case, numbers and special characters
- ◆ Something you can remember
- ◆ Techniques
 - *License plating* (becoming weaker)
 - Acronyms from phrases
 - Passphrases

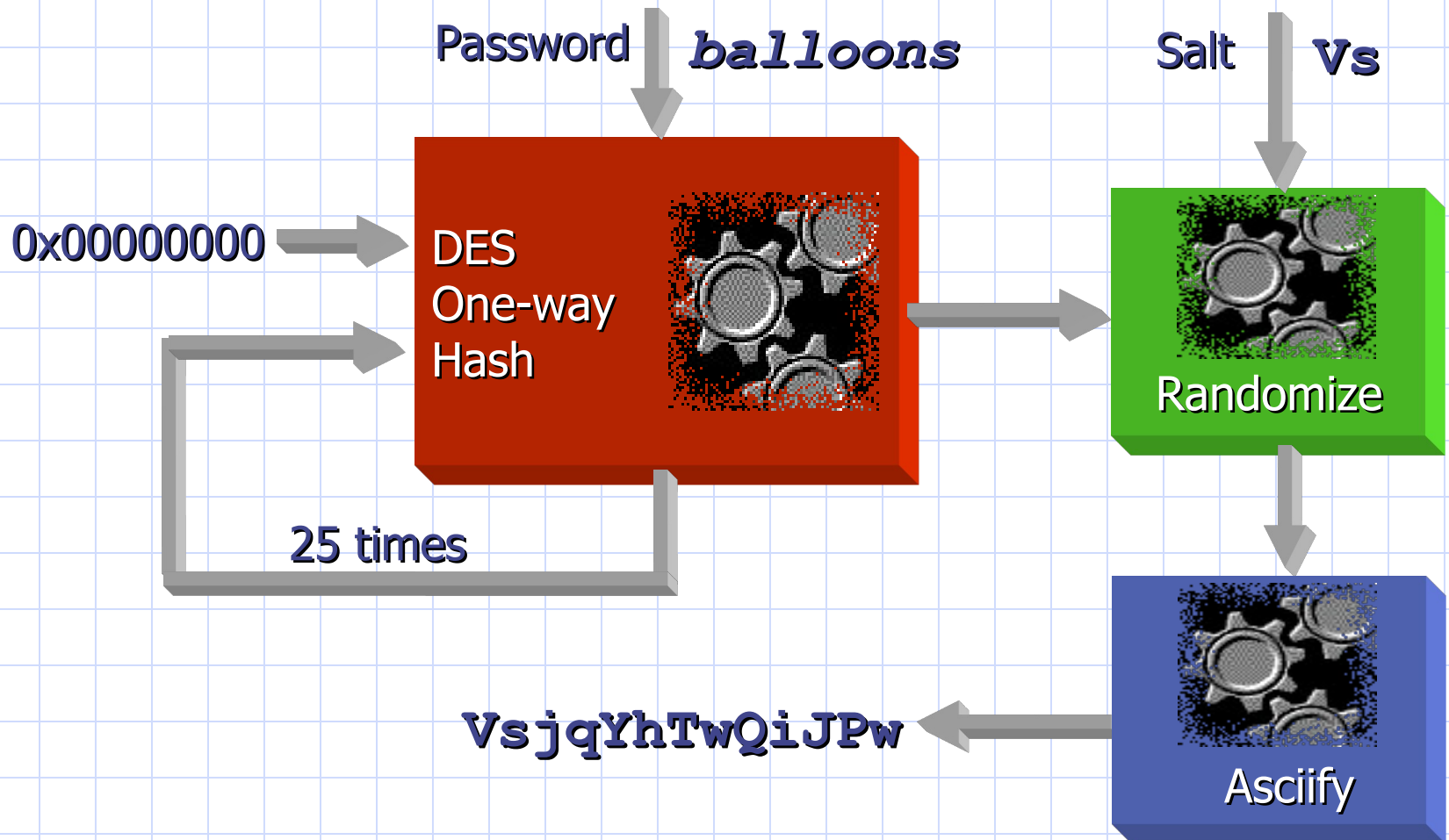
Pass Phrase Examples

- smo11012006
 - ◆ Poor
- ardl79BEf76357
 - ◆ 14 spaces
 - ◆ Hard to remember
 - ◆ Good
- MydogSkiplovestoplayfetcheveryday
 - ◆ 33 spaces
 - ◆ Easier to remember
 - ◆ Better
- Myd0gSkipluvs2playfetchev3ryday
 - ◆ 31 spaces
 - ◆ Easier to remember
 - ◆ Limit Duplicate letters –substitute with numbers, punctuation, or special characters
 - ◆ Best

UNIX Password Storage

- ◆ Uses a One-Way Hash Encryption
 - Based on DES
 - Uses a 2 character "salt"
 - MD5/SHA1 – More secure replacement
- ◆ '*' Used to lock accounts
- ◆ Readability Issue

UNIX Password Encryption



Newer systems use MD5/SHA1 now

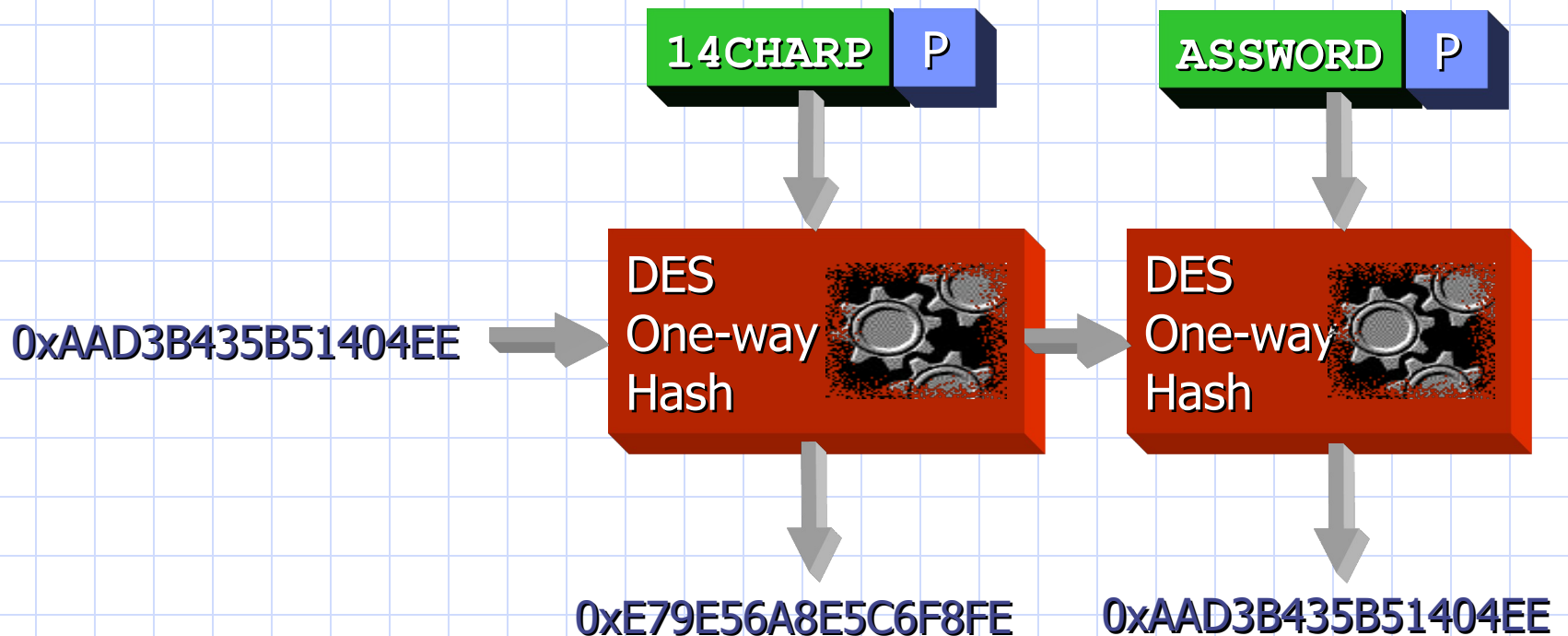
Valid: A-Za-z0-9./

/etc/shadow

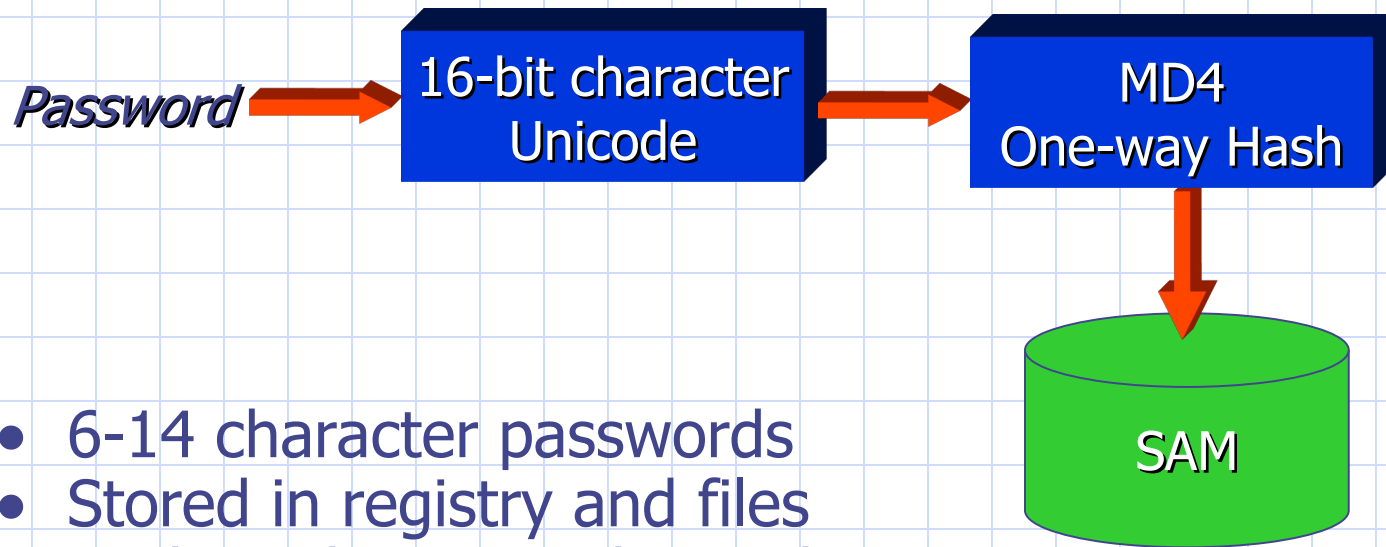
- ◆ Password field in `/etc/passwd` replaced with 'x'
- ◆ Readable only by superuser
- ◆ Contents
 - Username
 - Password
 - Password aging information

Lan Manager Password Encryption

Password *14charpassword*



Windows Password Encryption



- 6-14 character passwords
- Stored in registry and files
- Backwards compatible with LAN Manager (2nd entry)

Special Users

- ◆ Guest
- ◆ System Accounts
- ◆ *Superuser / Administrator*
 - Full Access to all system resources
 - *Superuser* Equivalency

"Principle of Least Privilege"

Linux System Accounts

◆ root

- System Administration account
- UID of 0

◆ bin – Owner of standard system programs

◆ daemon – Owner of (most) system daemons

◆ mail – Owner of mail system

◆ nobody – Unprivileged system account

Linux Groups

◆ Stored in `/etc/group`

- Group Name
- Password (rarely used)
- Group ID Number (GID)
- List of members

◆ `newgrp` – Change default group

```
sysadmin:*:14:root,sorr
```

Common Windows Groups

- ◆ Administrators
- ◆ Backup Operators
- ◆ Power Users
- ◆ Network Configuration Operators
- ◆ Remote Desktop Users
- ◆ Users

Role Based Model

Linux User Account Creation

- ◆ Add entry to `/etc/passwd`
- ◆ Create initial password (`/etc/shadow`)
- ◆ Add entry to `/etc/group` (optional)
- ◆ Create home directory & copy startup files into it
- ◆ Create mail file (optional)
- ◆ Test!!!

Linux Account Creation Tools

◆ Manual Creation

- `vipw / vigr`
- `passwd user`
- Default scripts found in `/etc/skel`

◆ `useradd`

◆ GUI Tools

- Linux Web-based tools (e.g. Webmin)
- Windows: Computer->Manage->Local Users and groups

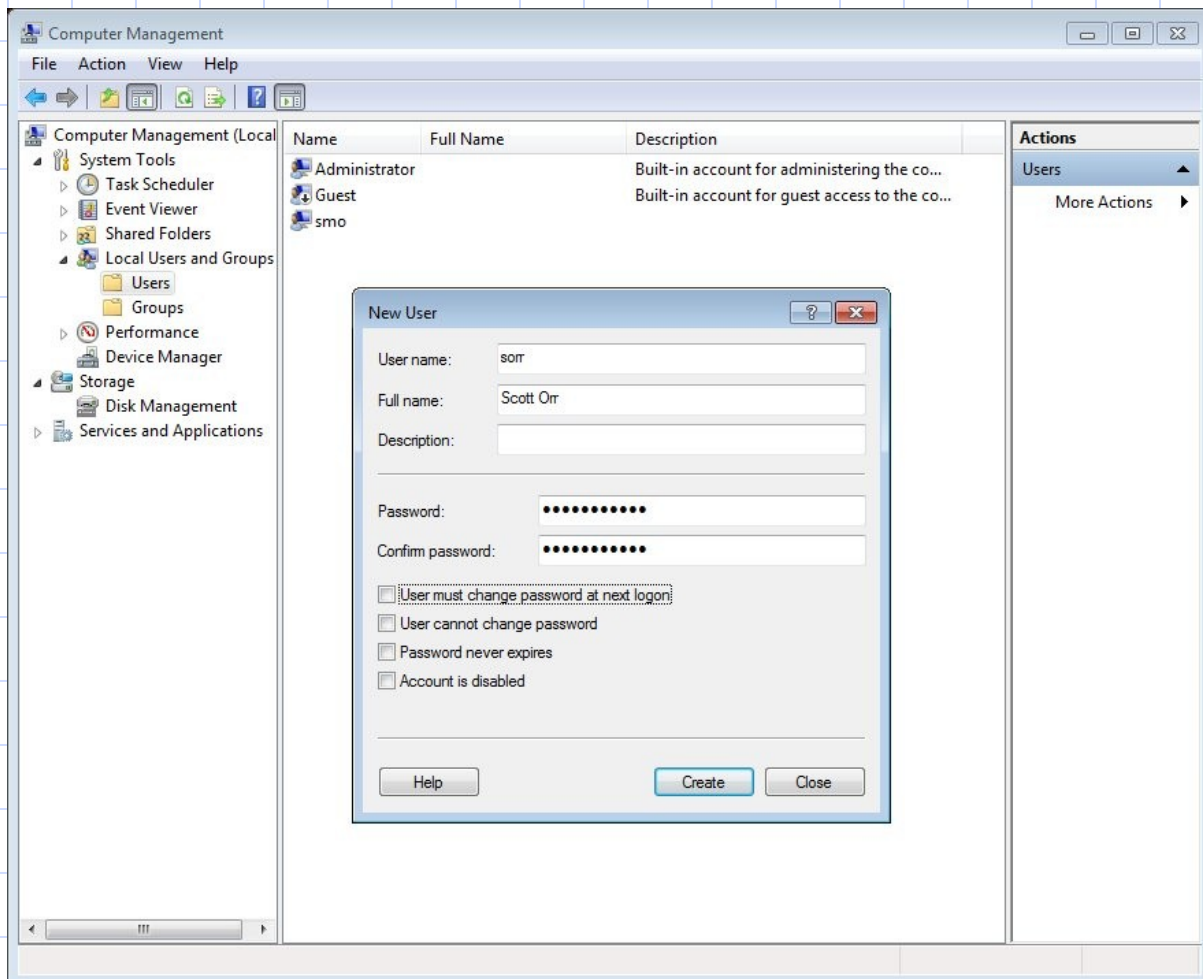
Linux Account Modification Tools

- ◆ **passwd** – Change password
- ◆ **chfn** – Change GECOS entry
- ◆ **chsh** – Change default shell
(Listed in **/etc/shells**)
- ◆ **usermod** & GUI tools
- ◆ Disabling accounts
 - Locking password ('*' as first character)
 - Change shell to **/bin/nologin**

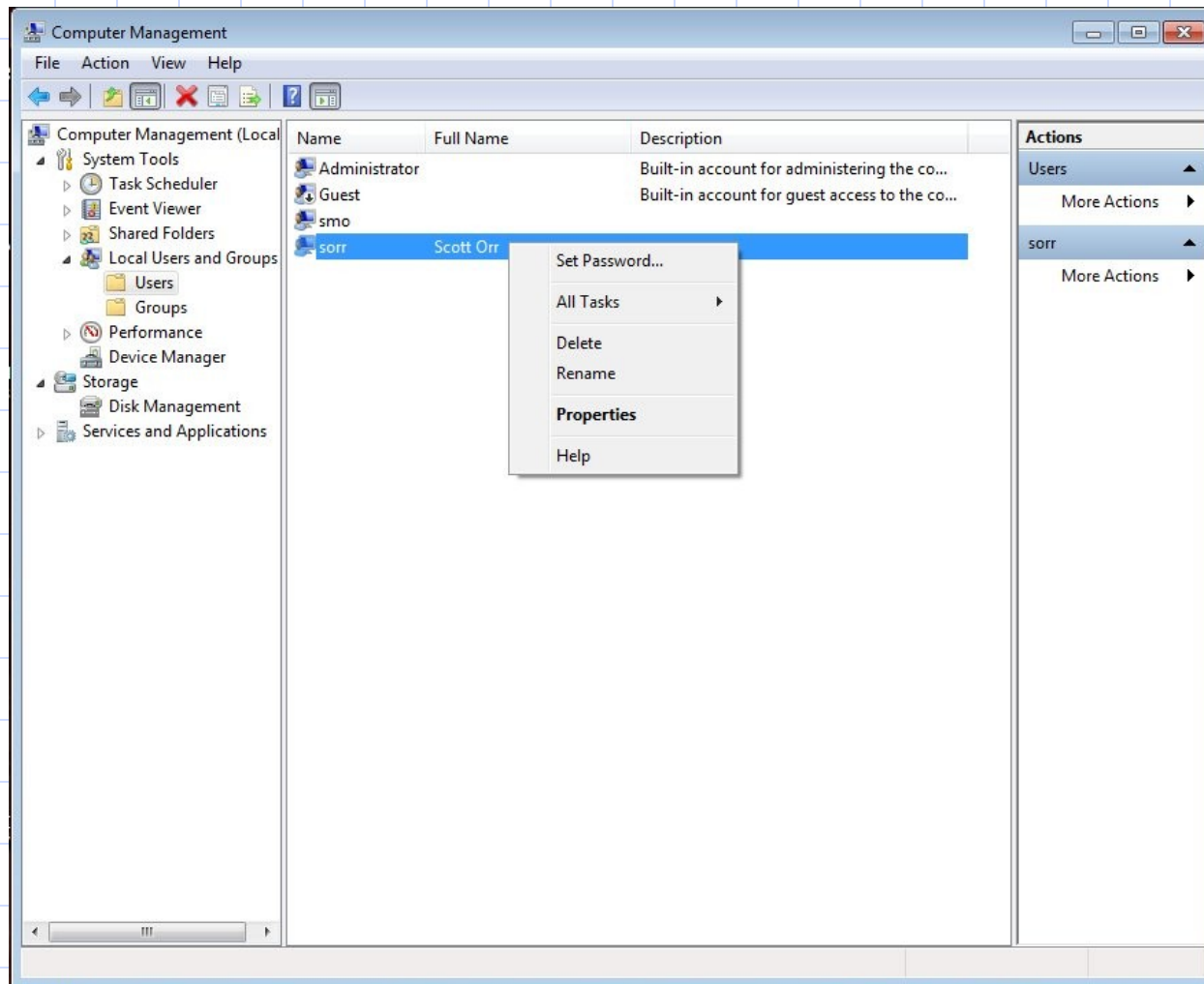
Removing a Linux Account

- ◆ Kill any processes owned by user
- ◆ Remove all files owned by user
- ◆ Remove account entries (Linux)
 - `/etc/passwd`
 - `/etc/shadow`
 - `/etc/group`
- ◆ `userdel` & GUI tools

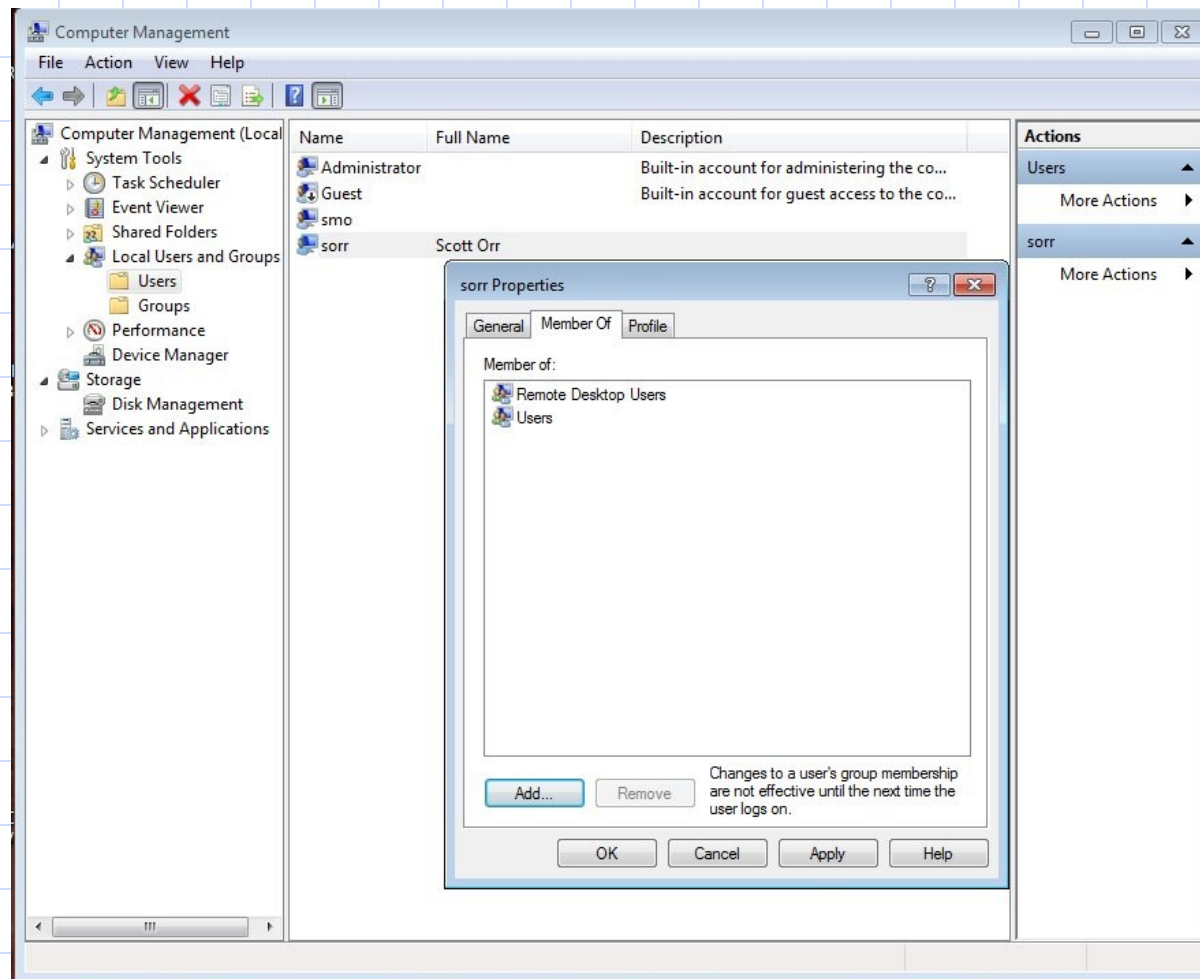
Adding a Windows Account



Modify a Windows Account



Windows Group Membership



Windows User Mgmt CLI

◆ List local users

- `net user`

◆ Create a new user

- `net user <username> <password>
/add`

◆ Delete a user

- `net user <username> /del`

Windows Group Mgmt CLI

◆ List groups

- `net localgroup`

◆ List group membership

- `net localgroup <groupname>`

◆ Add/Remove group members

- `net localgroup <groupname>
 <username ...> </add | /del>`